



Vehicle Service Booking and Tracking System

Members: M Shashanth – 2520090129, M Akshay – 2520090065, G Nandha kishore – 2520080050 | Guide: Dr. M Gayatri

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Abstract

- Web-based platform for vehicle service management: customers register vehicles, book services and track status; service centers manage bookings, job cards, technicians, spare parts and invoices.
- Goal: reduce manual work, improve service visibility.

Problem Statement & Objectives

- Problem:** Manual booking, hard-to-track service, paper-based records and billing, difficult technician and spare-part management.
- Objective 1:** Online service booking and vehicle management
- Objective 2:** Job cards, technician allocation, status tracking
- Objective 3:** Spare parts, digital invoices and notifications
- Scope:** Customers and service centers; booking, job-card, invoice and payment modules

Existing vs Proposed System

Feature	Existing System	Proposed System
Booking	Phone / visit	Online booking
Job cards	Manual job card	Digital job cards
Billing	Manual, paper-based	Digital invoices
Tracking	Limited visibility	Status tracking
Limitation / Gap	Delayed updates	Notifications

Feasibility Study

- Technical:** React.js, Node.js, Express.js and MySQL with REST APIs; built and run in VS Code
- Economic:** Open-source stack: React.js, Node.js, Express.js and MySQL
- Operational:** Simple interface, easy booking and clear status tracking
- Schedule:** Completion within the semester timeline

Results & Testing

- Postman API testing: requests, authentication, status codes
- Data validation, transactions, indexed queries

Conclusion

- Centralized vehicle service management with faster, easier booking and better status visibility
- Digital job cards and invoices with a secure REST-based backend

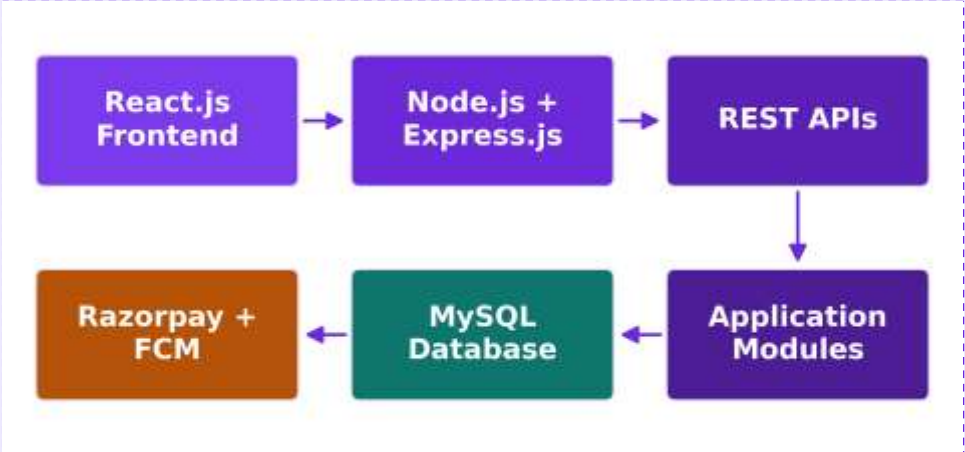
Future Scope

- Microservices and Kubernetes
- Redis and MongoDB
- Real-time tracking and vector search

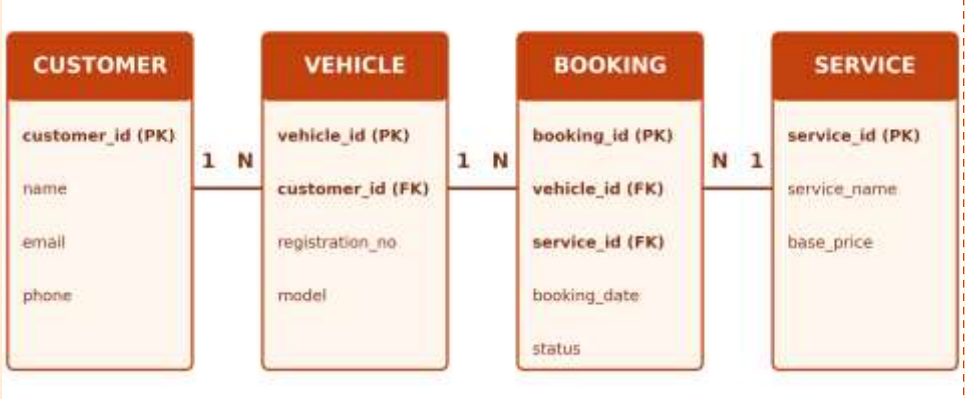
Tools & Technologies

- Front-End:** React.js
- Back-End:** Node.js + Express.js + REST APIs
- Database:** MySQL
- Tools:** Razorpay, FCM, Postman, VS Code

System Architecture



ER Diagram & Database Design



Implementation Screenshots

